

GFL Environmental Inc.

Data Engineer — Take-Home Assignment

Commercial Division: Route Profitability & Fleet Efficiency

Time	4 – 6 hours
Dataset	gfl_commercial_routes.csv — 12,000 records (attached)
Grain	One row per route per day. Each route runs 100+ stops/day.
Stack	PySpark + Delta Lake
Submission	GitHub repository link with README

The Business Problem

GFL's commercial division operates high-density front-load routes serving retail, restaurants, offices, healthcare, hospitality, and industrial customers. Each route runs 100 or more stops in a single day, spanning multiple customer segments and waste streams.

Leadership sees the top-line revenue but cannot easily answer: which routes, areas, and BUs are actually making money once fuel, labour, and disposal costs are fully accounted for? Some routes look high-revenue but bleed margin. Others quietly outperform.

Build a reliable, repeatable view of route-level profitability that can be sliced by region, BU, and area.

The Dataset

Each row represents one route on one day — already aggregated from individual stops.
Columns:

Column(s)	What it represents
route_date_key	Primary key — one row per route per day
date, year, month, quarter, day_of_week	Date dimensions
region, bu, area, route_id	Geography & operational hierarchy
primary_waste_stream	Dominant material — General Waste, Recycling, Organics, Cardboard
primary_customer_segment	Dominant segment — Retail, Restaurant, Office, Healthcare, etc.

num_drivers, num_trucks	Crew and fleet deployed on this route/day
total_stops, completed_stops, missed_stops	Stop counts (commercial routes run 100+ stops/day)
total_distance_km	Total km driven on the route
total_fuel_used_litres	Total fuel consumed (litres)
total_labour_hours	Total driver hours worked
total_yards, total_tonnes	Material collected
avg_revenue_per_stop	Avg billed revenue per completed stop
total_revenue	Gross revenue billed for this route/day
disposal_cost	Tipping / processing fees at disposal site
fuel_cost, labour_cost, maintenance_cost, admin_cost	Operating cost components
total_cost	Sum of all costs
net_revenue	total_revenue – disposal_cost
gross_profit, gross_margin_pct	Bottom-line profitability
scheduled_hours, actual_hours, delay_minutes, on_time_flag	Schedule adherence
incident_flag, incident_type	Safety events

What to Build

Part 1 — Delta Lake Pipeline

- Ingest the CSV into a Bronze Delta table with an enforced schema.
- Build a Silver layer with any derived metrics you believe matter for route profitability.
- Build a Gold aggregation table — your call on the right grain — partitioned to support efficient BI queries by region, BU, and area.
- Use at least one Delta feature meaningfully — **MERGE, time travel, OPTIMIZE/ZORDER, or schema evolution**. Explain why you chose it.

Part 2 — Route Profitability Analysis

- Define what underperforming means for a commercial route. There is no single right answer — show your reasoning.
- Identify the underperforming routes, areas, or BUs.
- Determine the primary cost drivers behind low-margin route-days.
- Quantify whether performance is improving or deteriorating across the 3-year window.

- One data-backed recommendation the operations team could act on next quarter.

Part 3 — Data Model

- Design a dimensional model to support a Route Profitability dashboard sliceable by date, region, BU, and area.
- Include an ERD and DDL. Note how Delta Lake's features influence your design choices.

Submission

- **GitHub repository** — public, with all code and the notebook
- **README** — setup, your metric decisions, and 3–5 key findings
- **AI usage** — tools used, for what, and where you corrected or overrode them

We evaluate judgment, not completeness. The data has the raw numbers. Deciding which metrics matter for commercial route profitability — and explaining why — is the core of this assessment.

Questions? Reply to this email. Good luck.